

15 Discrete Time Fourier Transform

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \\ x[n] &= \frac{1}{2\pi} \int_{\hat{\omega}=2\pi} X(e^{j\hat{\omega}}) e^{j\hat{\omega}n} d\hat{\omega} \end{aligned}$$

We have the discrete time Fourier transform as

$$\tilde{x}[n] = x[n] = \frac{1}{2\pi} \int_{2\pi} x(e^{j\hat{\omega}}) e^{j\hat{\omega}n} d\omega$$

(assuming that we go to really high n samples, so that it is a Riemann sum that we're saying is an integral

where

$$X(e^{j\hat{\omega}}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\hat{\omega}n}$$

and

$$a_k = \frac{1}{N} X(e^{jk\hat{\omega}_0})$$

This $X(e^{j\hat{\omega}})$ is continuous and periodic in 2π , so $X(e^{j\hat{\omega}}) = X(e^{j(\hat{\omega}+2\pi)})$

The DTFT converges for absolutely summable signals.

Note, our $X(e^{j\hat{\omega}})$ is a discrete signal for each input $\hat{\omega}$, but we have an infinite number of those $\hat{\omega}$ that we are considering. This is why we have to integrate over all of them.

Recall that if $|\alpha| < 1$ then $\sum_{n=0}^{\infty} \alpha^n = \frac{1}{1-\alpha}$

The magnitude of $a e^{-j\hat{\omega}} < 1$ since $|\alpha| < 1$ and $e^{-j\hat{\omega}} = 1$. This gives us nicely that

$$X(e^{j\hat{\omega}}) = \frac{1}{1 - a e^{-j\hat{\omega}}}$$

Using Eulers identity $e^{j\hat{\omega}} = \cos(\hat{\omega}) + j \sin(\hat{\omega})$, we have

$$X(e^{j\hat{\omega}}) = \frac{1}{(1 - a \cos(\hat{\omega}))^2 + a \sin(\hat{\omega})^2}$$

Ex 2

let $x[n] = \delta[n - n_0]$

$$\begin{aligned} X(e^{j\hat{\omega}}) &= \sum_{n=-\infty}^{\infty} \delta[n - n_0] e^{-j\hat{\omega}n} \\ &= e^{-j\hat{\omega}n_0} \end{aligned}$$

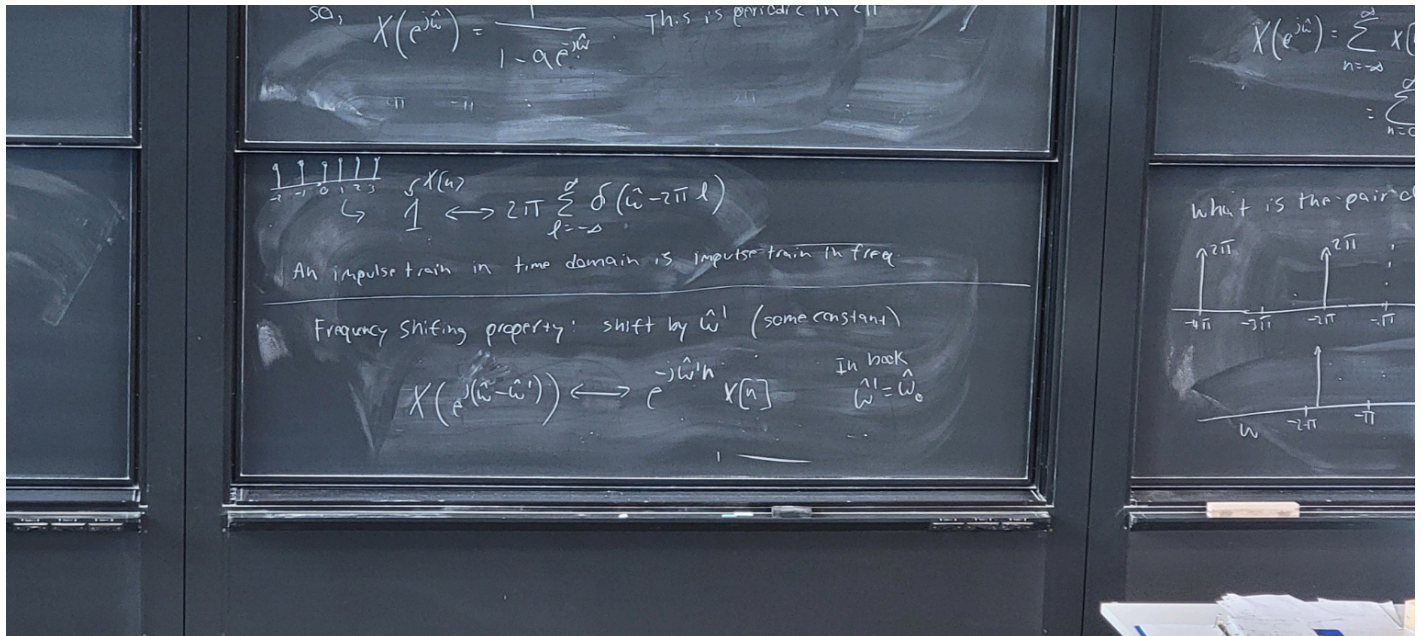
So the Fourier transform pair

$$\delta[n - n_0] \leftrightarrow e^{-j\hat{\omega}n_0}$$

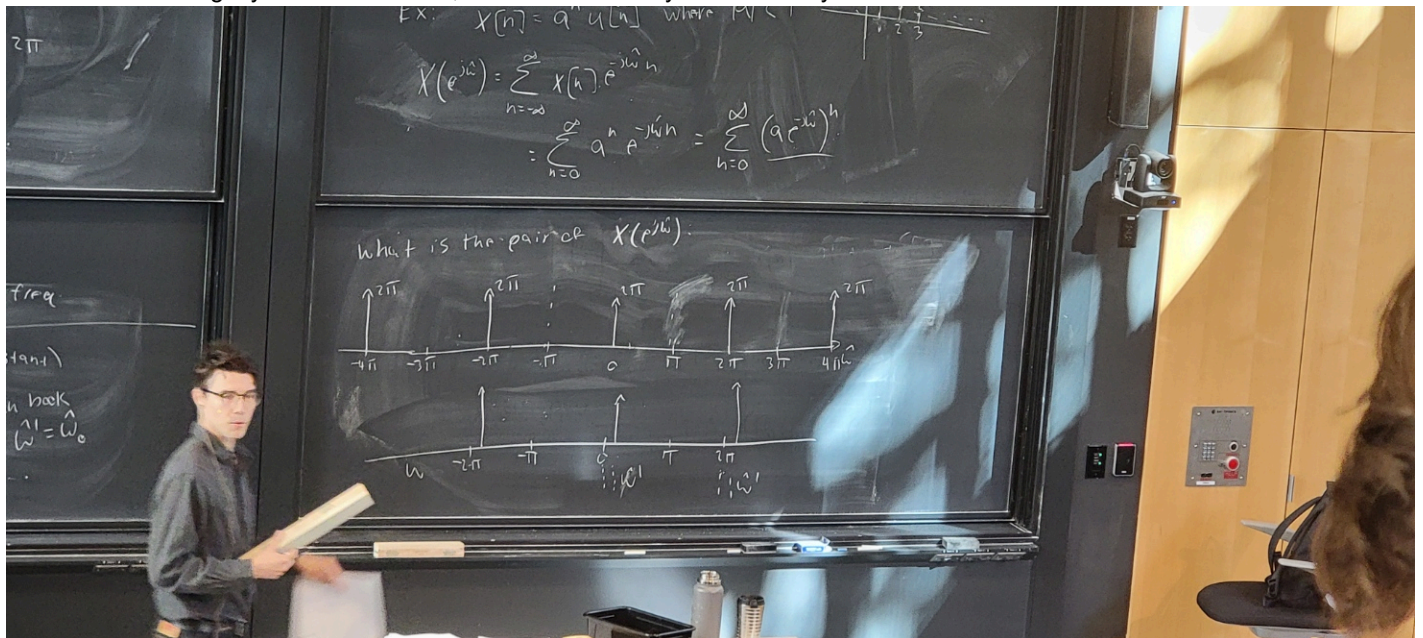
Ex 3

Lets take an $x[n]$ that is three spikes of height 1, at -1, 0, and 1. We could use the T formula, or we can use the pairs.

$$\begin{aligned}\delta[n-1] &\rightarrow e^{j\hat{\omega}} \\ \delta[n] &\rightarrow 1 \\ \delta[n+1] &\rightarrow e^{-j\hat{\omega}}\end{aligned}$$



Lets take a new slightly different function, that doesn't exactly land on every 2π .



$$2\pi \sum_{l=-\infty}^{\infty} \delta(\hat{\omega} - 2\pi l - \hat{\omega}_0) \leftrightarrow e^{-j\hat{\omega}_0 n}$$

If we have a periodic signal, using the Fourier series we would have

$$x[n] = \sum_{k=-\infty}^{\infty} a_k e^{jk\hat{\omega}_0 n}, \text{ where } \hat{\omega}_0 = \frac{2\pi}{N}$$